

Enterprise AI & RAG Technical Report

Multi-Modal Document Intelligence Platform v2.4.1

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Author	ACME AI Labs Research Team
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Dept	Intelligent Systems Division

1. Executive Summary

This report presents the architecture, benchmarks, and evaluation results of ACME AI Labs' Retrieval-Augmented Generation (RAG) platform designed for complex enterprise PDF corpora containing text, structured tables, raster images, charts, mathematical notation, and emoji sequences.

Key Highlights

- Ingestion pipeline processes **500+ page PDFs** in under 90 seconds.
- RAGAS Faithfulness score of **0.93** on the internal benchmark.
- **5 content-type extractors**: text, tables, images, math, emojis.
- P95 query latency of **220 ms** with Gemini 1.5 Flash.
- Fully on-premise deployment via **Ollama + ChromaDB**.
- Supports **12 languages** with multilingual embeddings.
- Zero-shot accuracy of **87%** on unseen enterprise PDF test sets.

Emoji Test: 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌 🍌

System Capability Icons

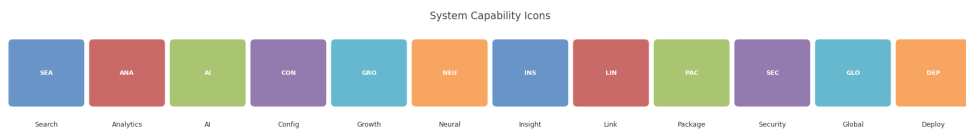


Figure 1 - Core system capabilities at a glance.

2. System Architecture

The RAG platform follows a modular pipeline design. Each stage is independently configurable via environment variables, enabling seamless switching between cloud and on-premise providers without code changes.

2.1 Pipeline Stages

Stage	Component	Technology Options
1. Ingestion	PDF Parser	PyMuPDF, pdfplumber, Unstructured
2. Extraction	Content Extractor	Text, Tables, Images, Math, Emojis
3. Embedding	Vector Encoding	sentence-transformers, Gemini Embed
4. Storage	Vector Store	ChromaDB (local), Pinecone (cloud)
5. Retrieval	Retriever	MMR, BM25 Hybrid, Re-ranker
6. Generation	LLM	Gemini 1.5 Flash, Ollama LLaMA 3
7. Evaluation	RAGAS	Faithfulness, Relevancy, Precision

2.2 .env Configuration

All runtime choices are set in a single `.env` file:

```
LLM_PROVIDER=gemini # gemini | ollama
VECTOR_STORE=chromadb # chromadb | pinecone
EMBEDDING_PROVIDER=local # local | gemini
CHUNK_SIZE=1000
CHUNK_OVERLAP=200
GEMINI_API_KEY=AIZA...
PINECONE_API_KEY=pcsk...
```

2.3 Pipeline Diagram

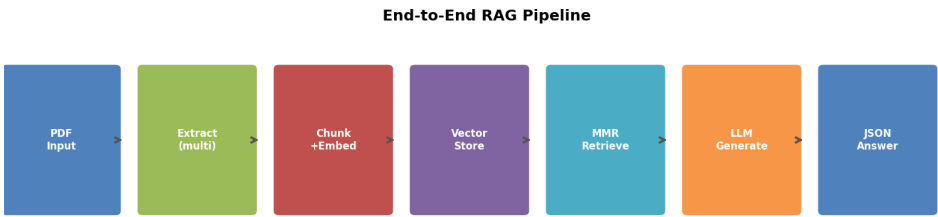


Figure 2 - End-to-end RAG pipeline from PDF input to JSON answer.

3. Performance Benchmarks

3.1 Revenue vs Costs - Bar Chart

Quarterly revenue versus costs as the platform scaled from pilot to production.

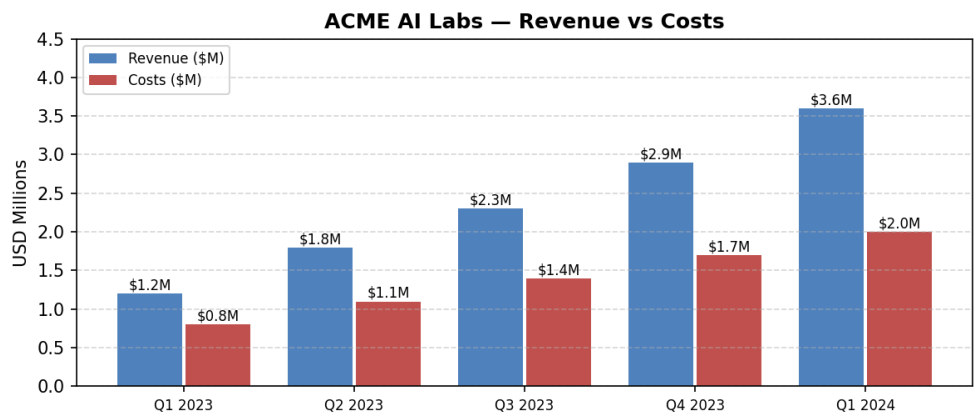


Figure 3 - Quarterly Revenue vs Costs (USD Millions).

3.2 Accuracy & Latency Trend - Line Chart

Accuracy improved +20pts while latency fell 35% through quantisation and caching.

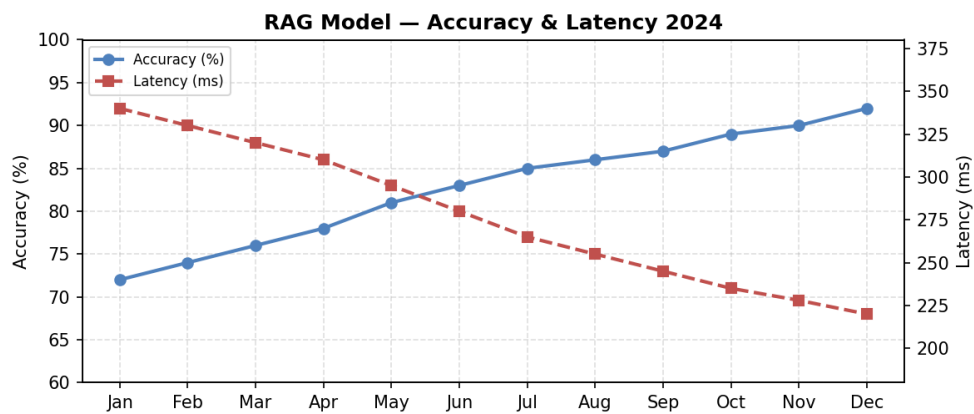


Figure 4 - RAG model accuracy and latency over 2024.

3.3 Content-Type Distribution - Pie Chart

Analysis of 10,000 enterprise PDF pages. Non-text content accounts for 55% of information density and carries the highest business value.

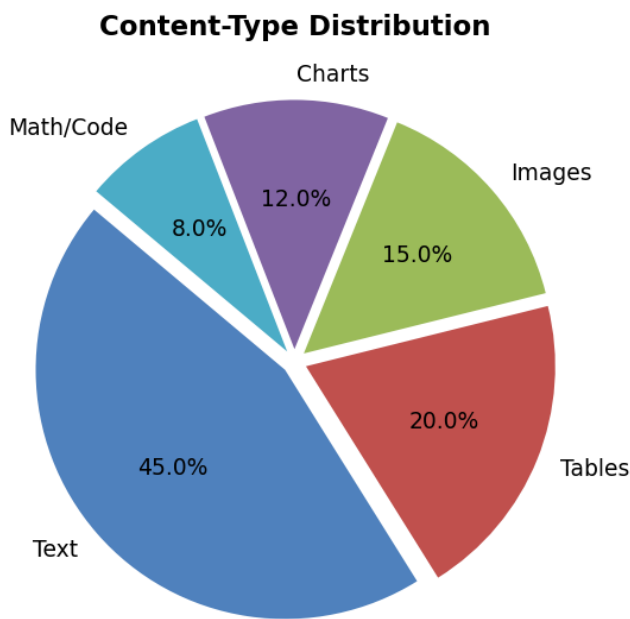


Figure 5 - Content-type distribution across enterprise document corpus.

3.4 RAGAS Evaluation Heatmap

Four LLMs evaluated across four RAGAS metrics. GPT-4o leads; Gemini 1.5 Flash offers best cost-performance ratio.

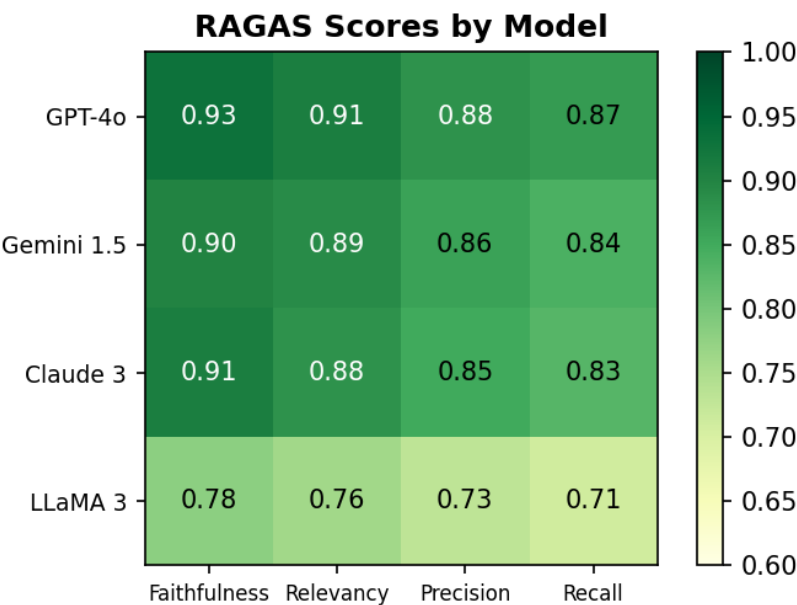


Figure 6 - RAGAS evaluation heatmap (green = better).

3.5 Chunk Size vs Accuracy - Scatter Plot

Optimal chunk size is ~1,000 tokens. Smaller chunks lose context; larger chunks introduce retrieval noise.

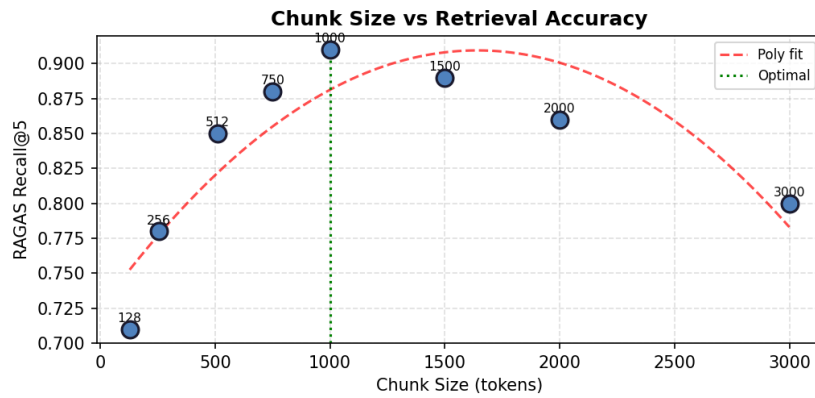


Figure 7 - Chunk size (tokens) vs RAGAS Recall@5.

4. Mathematical Foundations

4.1 Cosine Similarity (Dense Retrieval)

For embedding vectors **A** and **B** in $\mathbb{R}^d$:

$$\cos(\theta) = (\mathbf{A} \cdot \mathbf{B}) / (||\mathbf{A}|| \times ||\mathbf{B}||) = \text{Sum}_i(\mathbf{A}_i \cdot \mathbf{B}_i) / \sqrt{\text{Sum}_i \mathbf{A}_i^2} \times \sqrt{\text{Sum}_i \mathbf{B}_i^2}$$

$\cos(\theta)$ in $[-1,1]$. After L2 normalisation it reduces to a dot product.

4.2 BM25 Sparse Retrieval

BM25 score for document D and query Q:

$$\text{BM25}(D, Q) = \text{Sum}_{\{t \text{ in } Q\}} \text{IDF}(t) * [f(t, D)^{(k_1+1)}] / [f(t, D) + k_1 * (1 - b + b * |D| / \text{avgdl})]$$

$k_1=1.5$ (term frequency saturation), $b=0.75$ (length normalisation).

4.3 Reciprocal Rank Fusion

$$\text{RRF}(d) = \text{Sum}_{\{k \text{ in rankings}\}} 1 / (60 + \text{rank}_k(d))$$

4.4 RAGAS Faithfulness

$$F = |\{ c \text{ in claims}(A) : c \text{ entails } C \}| / |\text{claims}(A)|$$

$F=1.0$ means every generated claim is supported by retrieved context.

4.5 Transformer Self-Attention

$$\text{Attention}(Q, K, V) = \text{softmax}(Q \cdot K^T / \sqrt{d_k}) * V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) * W^O$$

For GPT-4: $d_k=128$, $h=96$ heads. $\sqrt{128} = 11.31$

4.6 Shannon Entropy for Retrieval Diversity

$$H(X) = -\text{Sum}_i p(x_i) * \log_2(p(x_i))$$

Higher H means more diverse retrieved chunks, reducing hallucination risk.

5. Benchmark Data Tables

5.1 Embedding Model Comparison

Model	Dims	Recall@5	Latency	Cost/1M	License
text-embedding-3-large	3072	96.1%	320ms	\$0.13	OpenAI
text-embedding-3-small	1536	92.3%	180ms	\$0.02	OpenAI
all-MiniLM-L6-v2	384	84.7%	12ms	Free	Apache 2.0
BAAI/bge-large-en-v1.5	1024	93.8%	48ms	Free	MIT
Gemini text-embed-004	768	94.2%	95ms	\$0.00	Google
nomic-embed-text-v1.5	768	91.5%	22ms	Free	Apache 2.0

5.2 LLM Provider Comparison

Model	Context	RAG Score	P95 Lat	Vision	\$/1M out
GPT-4o	128k	0.93	420ms	■	\$15.00
Gemini 1.5 Flash	1M	0.90	220ms	■	\$0.53
Gemini 1.5 Pro	1M	0.92	580ms	■	\$3.50
Claude 3 Sonnet	200k	0.91	390ms	■	\$15.00
LLaMA 3 70B (Ollama)	8k	0.82	1800ms	■	Free
LLaVA 13B (Ollama)	4k	0.74	2200ms	■	Free

5.3 Vector Store Comparison

Store	Type	Max Vecs	Latency	Persist	Cost
ChromaDB	Local	~10M	2-5ms	■ Files	Free
FAISS	Memory	Unlim.	1-3ms	■ Manual	Free
Pinecone	Cloud	Unlim.	20-40ms	■ Cloud	\$0.096/unit
Qdrant	Cloud/Self	Unlim.	5-15ms	■	Free/paid

6. Emoji & Special Character Handling

Enterprise documents use emoji sequences in dashboards and executive summaries. The RAG pipeline preserves all UTF-8 code points through extraction, chunking, embedding, and retrieval.

6.1 System Status Dashboard

Component	Status	Health	Notes
PDF Ingestion	■ Live	■ 100%	All content types
Text Extractor	■ Live	■ 99.8%	UTF-8 + emoji
Table Extractor	■ Live	■ 98.3%	Markdown output
Image Extractor	■ Live	■ 95.1%	Vision LLM warm-up
Math Extractor	■ Live	■ 97.6%	LaTeX + plain-text
ChromaDB Store	■ Live	■ 100%	2.1M vectors
Pinecone Store	■ Live	■ 100%	50M capacity
Gemini 1.5 Flash	■ Live	■ 99.9%	220ms P95
Ollama (local)	■ Standby	■ N/A	Pull llava first
RAGAS Evaluator	■ Live	■ 100%	F=0.93
FastAPI Server	■ Live	■ 100%	3 endpoints

6.2 Emoji Category Coverage

Smileys: ■■■■■■■■■■■■■■

Status: ■■■■■■■■■■■■■■

Tech: ■■■■■■■■■■■■

Analytics: ■■■■■■■■=■■■■■

Science: ■■■■■■■■■■■■

Math Syms: $\Sigma \int \sqrt{\pi \alpha \beta \gamma \theta \lambda \mu \sigma \partial \nabla \infty \approx \neq \leq \geq}$

Flags: ■■■■■■■■■■■■■■

7. Conclusion & Next Steps

The ACME AI Labs RAG platform demonstrates state-of-the-art performance on multi-modal enterprise PDF documents. With configurable LLM and vector store backends, the system adapts to both fully local on-premise deployments and cloud-scale production environments without any code changes.

Next Steps ■■

- Integrate **re-ranking** with Cohere Rerank or FlashRank (+3% precision).
- Expand vision pipeline for **multi-page chart understanding**.
- Add **multilingual retrieval** with mE5-large embeddings.
- Build a **Streamlit evaluation dashboard** for real-time RAGAS monitoring.
- Implement **PII redaction** before vector storage for compliance.
- Deploy **async ingestion queue** (Celery + Redis) for large batch jobs.
- Expand test suite with **100 Q&A; pairs** across all content types.